MILITARY SPECIFICATION

ANTIMONY TRIOXIDE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of antimony trioxide (Sb_2O_3).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572 - Wood-Preservative; Water-Repellent.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical Society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. Antimony trioxide shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as Sb_2O_3)	99.0	----	4.2.4.1
Arsenic (As)	----	0.20	4.2.4.2
Chloride (Cl)	----	0.005	4.2.4.3
Heavy metals (as Pb)	----	0.050	4.2.4.4
Iron (Fe)	----	0.010	4.2.4.5
Sulfate (SO_4)	----	0.010	4.2.4.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the antimony trioxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/4 pound (lb) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Fiberboard fillers and pads, strapping, or tape not as specified.
(f) Wood preservation incorrect or missing (level A packing only).
(g) Marking incorrect, missing, or illegible.

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/4 lb. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as Sb_2O_3). Weigh to the nearest 0.1 milligram (mg) approximately 0.2 gram (g) of the specimen into a 125-milliliter (ml) glass-stoppered flask. Dissolve the specimen by adding 10 ml of 10-percent tartaric acid solution and a few drops of concentrated hydrochloric acid and warming the solution. Cool to room temperature and add anhydrous sodium carbonate until the solution is nearly neutral. Then add 40 ml of cold, saturated sodium bicarbonate solution and 5 ml of starch indicator solution and immediately titrate with 0.1N iodine solution. Calculate the percent antimony trioxide as follows:

$$\text{Percent antimony trioxide} = \frac{7.29\ AB}{W}$$

where: A = Milliliters of iodine solution used,
B = Normality of iodine solution, and
W = Weight of specimen in grams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

4.2.4.2 Arsenic. Weigh 0.100 g of the specimen and dissolve in 10 ml of concentrated hydrochloric acid and add 10 ml of stannous chloride solution (40 g of stannous chloride dissolved in 60 ml of concentrated hydrochloric acid). Any color produced shall be no darker than that produced in a control in which 0.20 mg of arsenic ion is tested in the same manner as the specimen.

4.2.4.3 Chloride. Weigh 0.400 g of the specimen and boil with 20 ml of 20-percent tartaric acid for a few minutes. Filter and add 1 ml of concentrated nitric acid and 1 ml of 1.7-percent silver nitrate solution to the filtrate. Any turbidity produced shall be no greater than that produced in a control containing 0.02 mg of chloride ion and the quantities of reagents used in the specimen test in the same final volume as the specimen.

4.2.4.4 Heavy metals (as Pb). Weigh 1.000 g of the specimen and dissolve in 3 ml of concentrated hydrochloric acid. Add 5 g of tartaric acid and dilute to 45 ml with water. Pour this solution into a mixture of 150 ml of water and 150 ml of a solution containing 12 g of sodium sulfide, nonahydrate and 8 g of sodium hydroxide dissolved in water and diluted to 200 ml with water. Warm the solution and allow the insoluble sulfides to settle. Filter, wash thoroughly with hydrogen sulfide water, and discard the filtrate. Dissolve the precipitate remaining on the filter with 10 ml of hot 10-percent hydrochloric acid and dilute to 250 ml with water. Neutralize a 25-ml aliquot of the solution with 1N sodium hydroxide solution. Add 1 ml of 1N acetic acid and 10 ml of hydrogen sulfide water. Any brown color produced shall be no darker than that produced in a control containing 0.05 mg of lead ion, 1 ml of 1N acid, and 10 ml of hydrogen sulfide water and diluted to the same final volume as the specimen aliquot with water.

4.2.4.5 Iron. Weigh 1.000 g of the specimen and dissolve in 10 ml of concentrated hydrochloric acid. Slowly add 20 ml of water, filter if necessary, and add 30 to 50 mg of ammonium persulfate and 3 ml of 25-percent ammonium thiocyanate solution. Any red color produced shall be no darker than that produced in a control containing 0.1 mg of iron ion and the quantities of reagents used in the specimen test in the same final volume as the specimen.

4.2.4.6 Sulfate. Weigh to the nearest milligram approximately 10 g of the specimen and dissolve in the minimum amount of concentrated hydrochloric acid necessary. Carefully dilute to 150 ml with water, neutralize with ammonium hydroxide solution, and filter. To the filtrate, add 2 ml of concentrated hydrochloric acid, heat the solution to boiling, add 10 ml of 12-percent barium chloride solution, and allow to stand overnight. Filter any precipitate, wash thoroughly with water, and ignite to constant weight in a tared platinum cricible weighed to the nearest 0.1 mg. Cool to room temperature in a desiccator and weigh to the nearest 0.1 mg. Calculate the percent sulfate as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

$$\text{Percent sulfate} = \frac{41\ (A-B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One lb ± 0.1 ounce of antimony trioxide shall be packaged in a type I, class 1 glass bottle, with plastic screw cap, in inclosure b, as specified in PPP-C-301.

5.1.1.1 Intermediate packaging. Twelve bottles of antimony trioxide, packaged as specified in 5.1.1, shall be intermediately packaged as specified in PPP-C-301 except that only one tier of three rows having four bottles per row shall be placed in each intermediate container.

5.1.2 Level C. One lb ± 0.1 ounce of antimony trioxide shall be packaged as specified in 5.1.1 except that the bottle shall not be placed in inclosure b of PPP-C-301.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four of the intermediate packages specified in 5.1.1.1 shall be packed upright in a nailed wood, cleated plywood, or wirebound wood box as specified (see 6.2). The nailed wood box shall conform to class 2, style 4-1/2, for a type 2 load of PPP-B-621. The cleated plywood box shall conform to type overseas, style 1 or J, for a type 2 load of PPP-B-601. The wirebound box shall conform to class 3, style optional, for a type 2 load of PPP-B-585. The use of fiberboard fillers and pads shall be in accordance with PPP-C-301 to assure a tight pack. Each container shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification. In addition, unless specific provisions are made for preservation in the applicable box specification, each box or the finished, unpainted wood parts thereof shall be completely immersed for a minimum of 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.2 Level B. Antimony trioxide shall be packed as specified in 5.2.1 except that the box shall conform to style RSC, grade V3c of PPP-B-636, no preservative shall be required, and the box shall be reinforced by pressure sensitive adhesive filament reinforced tape as specified in the appendix to PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. Twelve bottles of antimony trioxide, packaged as specified in 5.1.2, shall be packed in a fiberboard box conforming to style RSC, type CF, class domestic, variety SW, grade optional of PPP-B-636. The box shall be fitted with fiberboard half-slotted separators which shall form close fitting cells for each of the bottles and which shall be the same height as the bottles. The inner faces of each box shall be lined with full box-size fiberboard pads. Additional pads shall be added where needed to assure a tight fit. Fiberboard pads and separators shall be of material conforming to grade W5c of PPP-F-320. The box shall be closed as specified for closure of class domestic boxes in PPP-B-636.

5.3 Marking. In addition to the marking specified in 5.3.1 and any additional marking required by the contract or order, unit, intermediate, and shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following special information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis or maximum limits of impurities.
(f) Net weight of contents.
(g) Precautionary marking:

WARNING !

Do not take internally.
Do not breathe dust.
Avoid skin contact.
Wash thoroughly before eating or smoking.

5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 using load type I except that strapping shall be restricted to galvanized steel of 1.25-inch minimum width and 0.035-inch minimum thickness. The load tie-down straps parallel to the 48-inch pallet dimension shall be modified to serve as load unitizing straps, containing the load but not the pallet. One load tie-down strap shall be added to, and centered between, those straps which are normally required to be parallel to the 40-inch pallet width. One horizontal strap shall be required for each tier of containers. The pallet shall conform to style 1, type optional, class 1, preserved of MIL-P-15011. All wood storage aids, if used, shall be preserved as specified for the pallet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Antimony trioxide is intended for use as a laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Kind of box required for level A packing (see 5.2.1).
(d) Additional marking, if required (see 5.3).
(e) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Air Force - 68

Preparing activity:

Army - MU(EA)

Project No. 6810-0461

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

TO DETACH FORM - CUT ON THIS LINE

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.